UV-Vis Band Gap Analysis — Figures

All figures for report inclusion (print-ready)

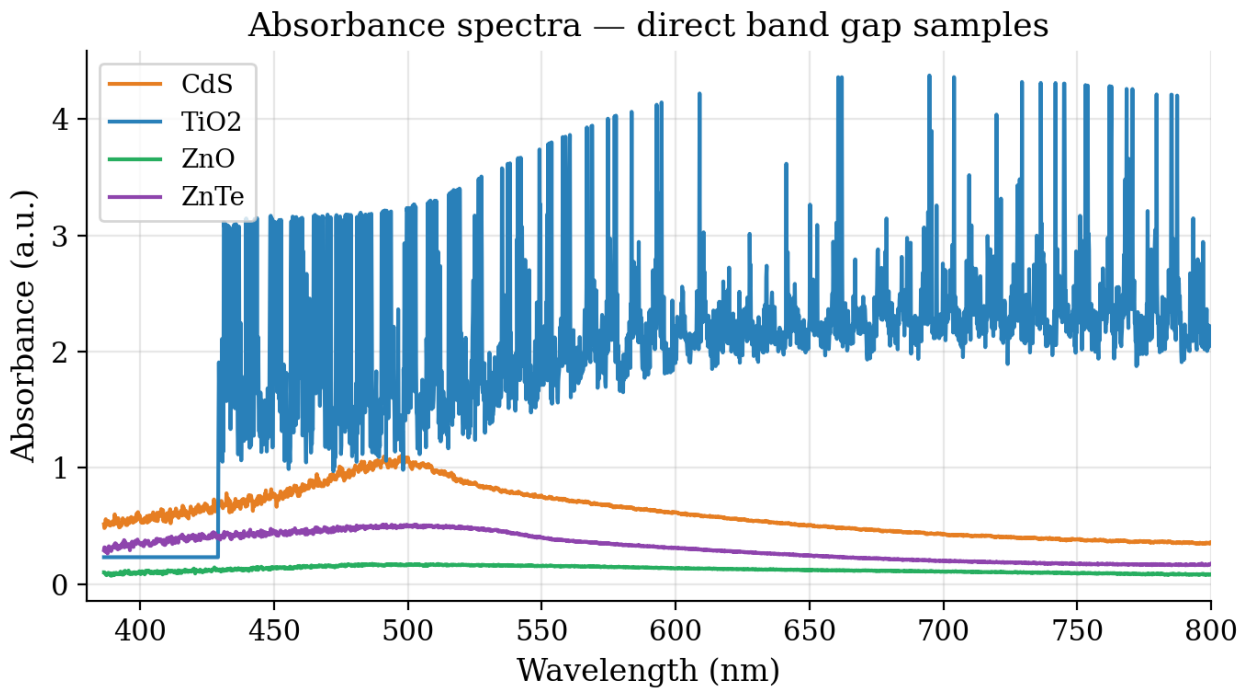

Figure 1. Absorbance vs. wavelength for all four direct band gap samples (CdS, TiO■, ZnO, ZnTe). ZnO and TiO■ absorb strongly at the short-wavelength detector limit (≈ 387 nm, 3.21 eV).

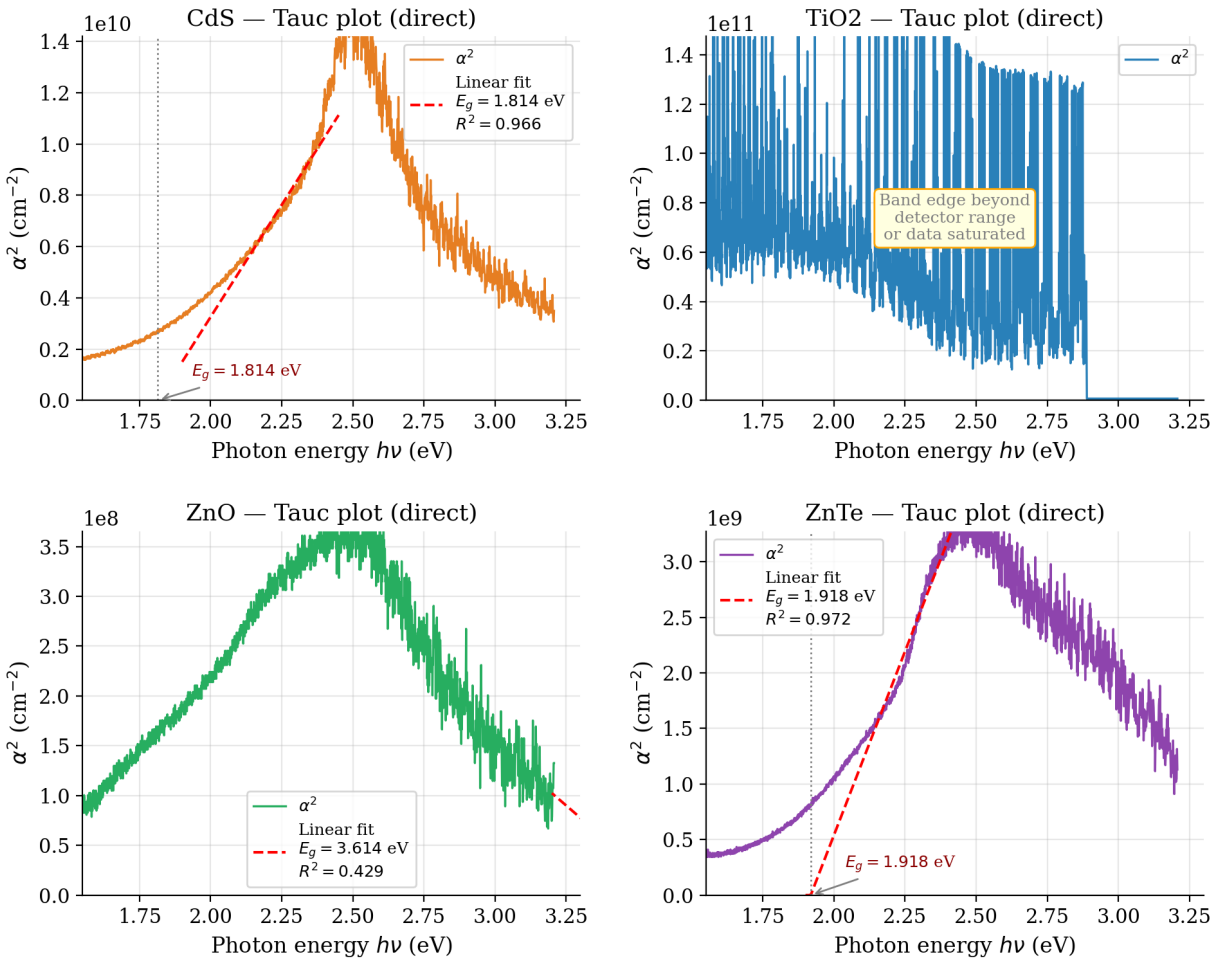

Figure 2. Tauc plots (α^2 vs. $h\nu$) for all four direct band gap samples arranged in a 2×2 grid. Red dashed lines show the linear fits; TiO■ data are saturated at the detector limit and no E■ is extracted.

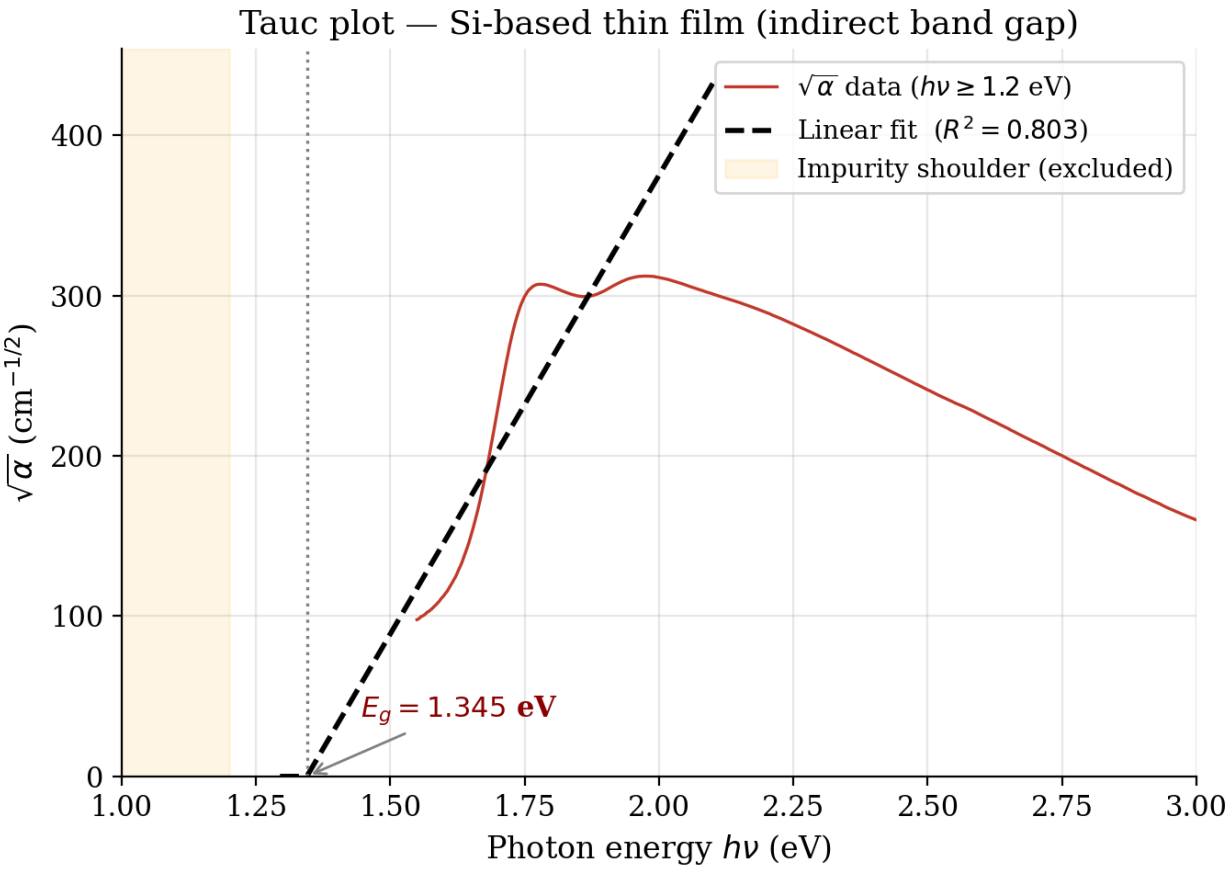

Figure 3. Tauc plot (√α vs. hν) for the indirect band gap Si-based polymer film (d = 150 nm). Orange shading marks the excluded impurity shoulder below 1.2 eV. Linear fit over 1.2–2.0 eV gives E■ = 1.345 eV (R^2 = 0.803).

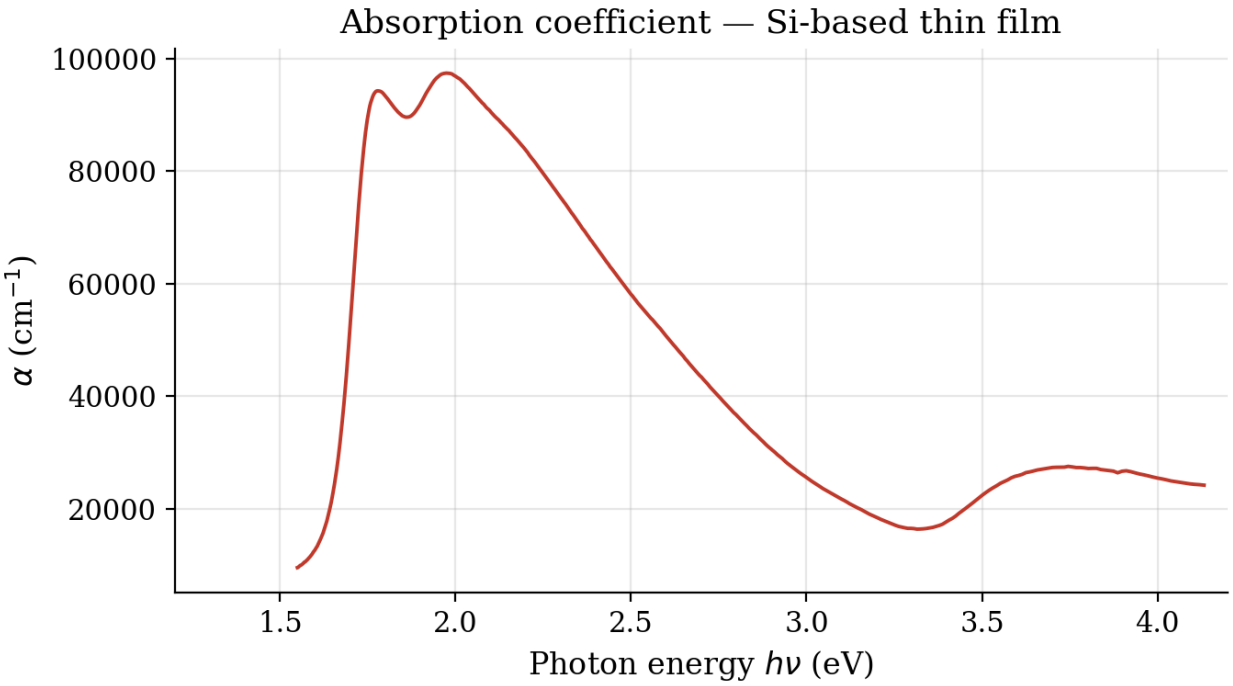

Figure 4. Absorption coefficient α vs. photon energy for the Si-based indirect band gap film. The gradual onset (no sharp edge) is characteristic of phonon-assisted indirect transitions.

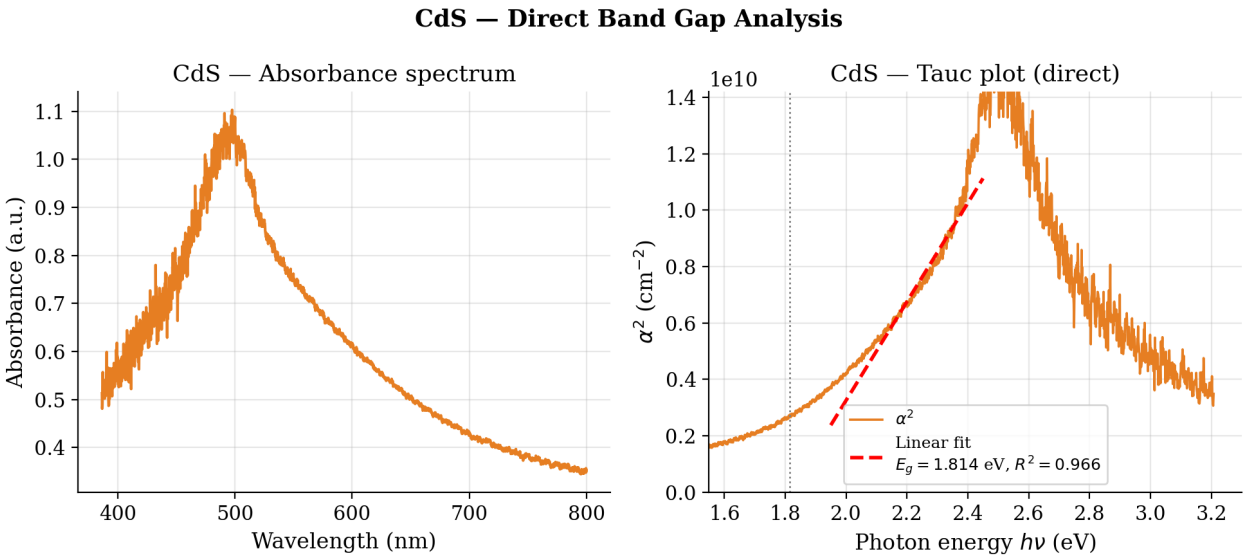

Figure 5. CdS thin film: absorbance spectrum (left) and Tauc plot with linear fit (right). E■(CdS) ≈ 1.81 eV; deviation from literature (2.42 eV) attributed to background absorption raising the α^2 baseline.

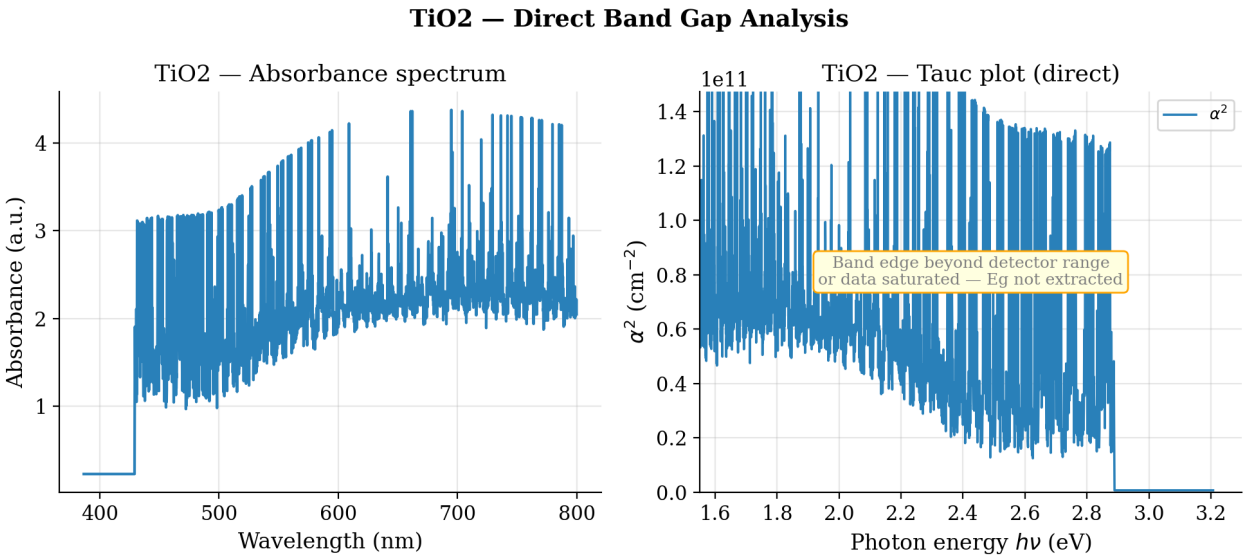

Figure 6. TiO■ thin film: absorbance spectrum (left) and Tauc plot (right). Absorbance saturates at A ≈ 0.23 above ~2.9 eV; the band gap (3.20 eV) lies at the spectrometer's short-wavelength limit and cannot be extracted reliably.

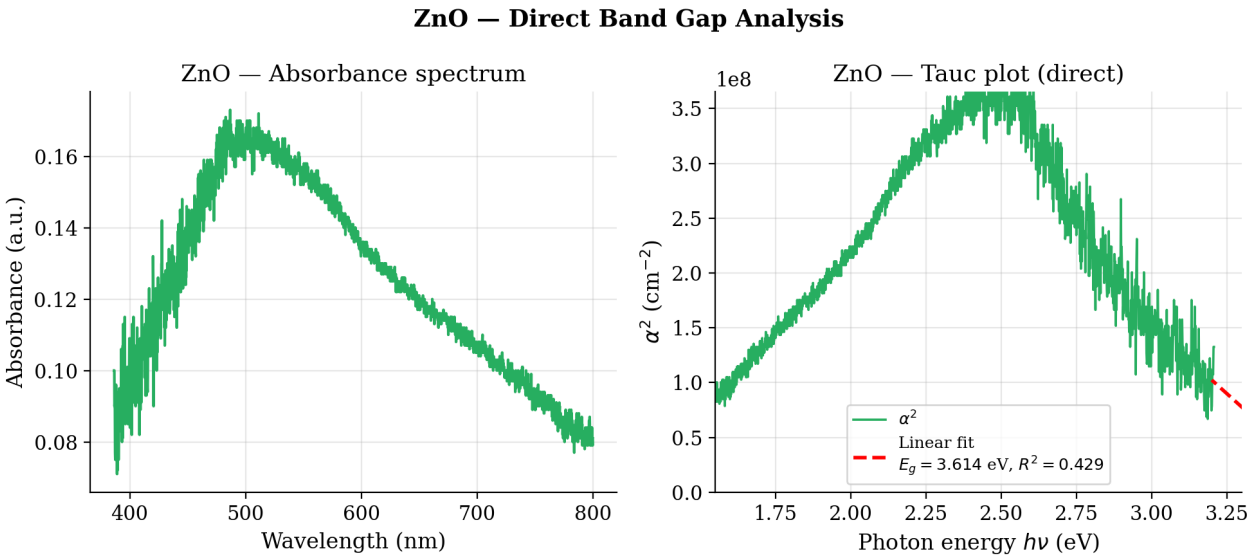

Figure 7. ZnO thin film: absorbance spectrum (left) and Tauc plot (right). Only a few data points lie near the band edge; E■(ZnO) ≈ 3.61 eV vs. literature 3.37 eV (+7 %).

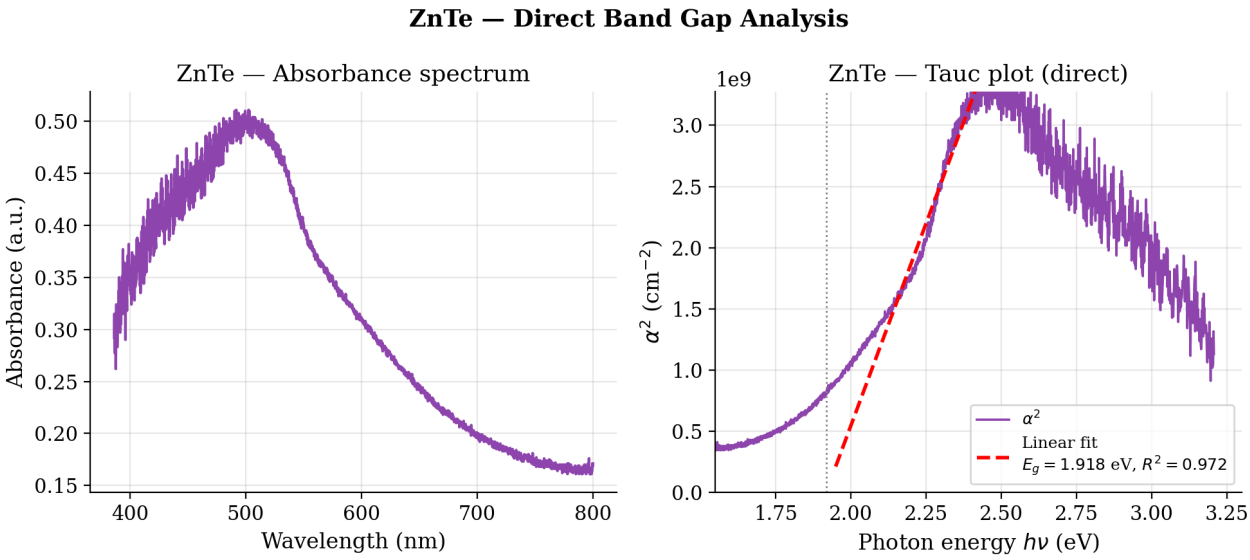

Figure 8. ZnTe thin film: absorbance spectrum (left) and Tauc plot (right). Cleanest onset of the four direct-gap samples; E■(ZnTe) ≈ 1.92 eV vs. literature 2.26 eV (–15 %).

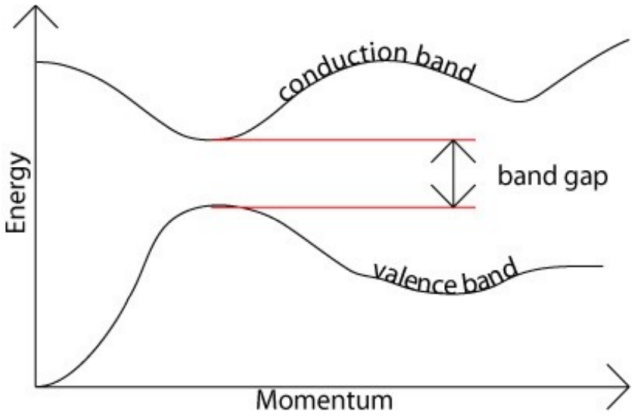

Figure 9. E–k schematic of a direct band gap semiconductor. The conduction band minimum (CBM) and valence band maximum (VBM) occur at the same crystal momentum k, so a single photon of energy hν ≥ E■ can directly create an electron–hole pair without phonon assistance. (Reproduced from lab manual.)

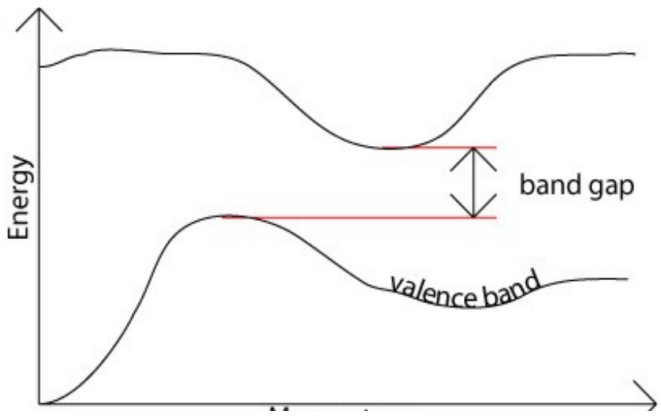

Figure 10. E–k schematic of an indirect band gap semiconductor. The CBM and VBM are offset in k-space; absorption requires simultaneous interaction with a phonon to conserve momentum, leading to a slower, gradual absorption onset compared with direct transitions. (Reproduced from lab manual.)